



A-POSTERIORI ERROR ESTIMATOR/CORRECTOR FOR NATURAL FREQUENCIES OF THIN PLATE VIBRATION PROBLEMS

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Abstract—From the finite element theory, a discretized finite element system approaches a continuous system if the finite element size approaches zero. Based on this fact, an asymptotic formula, which describes the asymptotic behaviour of the predicted natural frequency of a system from the finite element analysis, is presented in this paper. Since the proposed asymptotic formula expresses a relationship between the natural frequency and wave number of a thin plate system, it can be viewed as an *a-posteriori* error estimator/corrector and used to correct the error of the predicted natural frequency of the system from the finite element analysis. Owing to the use of some assumptions and a non-conforming thin plate bending element in the process of deriving the asymptotic formula, the corresponding error estimator/corrector can only result in a considerable, but not perfect, improvement on the predicted natural frequency from the finite element analysis when the thin plate is of finite size. An example has been given to show the possible application of the error estimator/corrector to the finite element analysis of thin plate vibration problems. Copyright © 1996 Elsevier Science Ltd.

1. INTRODUCTION

As is well known, exact solutions to the differential equation of a thin plate vibration problem are only available for the case of a rectangular plate with two opposite edges simply supported and with any conditions at the other two edges [1]. To calculate the natural frequencies of any other type of plate, some approximate methods have to be developed and employed. The early approximate method used for evaluating the natural frequencies of a rectangular thin plate with any boundary conditions was the Rayleigh–Ritz method [2]. In the Rayleigh–Ritz method, the conventional eigenvalue equation of a thin plate vibration problem was derived from considering the energy conservation of a system and minimizing the Rayleigh quotient which always resulted in higher natural frequencies than the true ones of the system. In fact, the Rayleigh–Ritz method provided upper bound natural frequencies for a thin plate vibration system, since the use of any incorrect trial function in the analysis is equivalent to the introduction of additional constraints to the system. The main disadvantage in using the Rayleigh–Ritz method is that it is often difficult to choose appropriate trial functions in the analysis, especially for a plate with complicated shape and boundary conditions.

The advent of modern computer technology and the finite element analysis technique [3, 4] provided a very powerful tool for solving thin plate vibration problems, since a thin plate with any complicated shape and boundary conditions, as well as any change of material properties, is easily modelled by the finite element method. Due to this reason, the development of thin plate bending elements has had extremely active attention since the concept of the finite element method began to become popular in the late 1950s. Most of the early developed thin plate bending elements were of the displacement type [5–7]. So far the number of available plate bending elements has become so large that it is difficult to choose an appropriate plate bending element in a finite element analysis, because no one particular plate bending element has been regarded as the so-called best element [8]. Unlike a plane stress element in two-dimensional elasticity, a plate bending element is C^1 element and accordingly, the conformity requirement for a thin plate bending element has been found to be particularly difficult because the interelement continuity is required for both the transverse displacement and the slope normal to the element boundary. If the above-mentioned requirement is satisfied in the process of constructing a thin plate bending element, the resulted element is called a conforming element and otherwise, a non-conforming element. It has been demonstrated that if any group of non-conforming elements represents rigid-body and constant curvature states exactly, the corresponding finite element solution can converge to the true value of strain energy [5], although the convergent pattern is not necessarily in monotonic form. Therefore, the non-conforming plate bending element, which violates the slope continuity between inter-element sides, has been still used in solving thin plate bending problems, especially when the following factors

such as simplicity of formulation, versatility of application, reliability, computational effort and accuracy are taken into account in the finite element analysis.

On the other hand, since the finite element method is an approximate method, the finite element solution cannot be safely used in engineering practice unless it is appropriately validated [9]. In this regard, the error estimate of the finite element solution has been given increasing attention by finite element analysts in recent years. However, the existing research work on this aspect is mainly limited to static problems [10, 11] and little, if any, work is on the error estimate for the natural frequencies of thin plate vibration problems in the finite element analysis. Therefore, the main purpose of this study is to develop an error estimator/corrector to correct the predicted natural frequency of a thin plate vibration problem by the finite element method and provide an improved solution to the same problem.

2. GOVERNING EQUATION AND FINITE ELEMENT FORMULATION OF A THIN PLATE VIBRATION PROBLEM

Based on the classical thin plate bending theory, a thin plate can be treated as a two-dimensional flat element, since both the length and the breadth of the thin plate are large enough in comparison to the depth of the plate. From the mechanical point of view, the thin plate can be viewed as the two-dimensional analogue of a beam, since the basic kinematics of the classical theory of thin plates is the same as that of Bernoulli–Euler beams. On the other hand, from the finite element analysis point of view, the thin plate can be considered as the analogue of a plane stress element in two-dimensional elasticity, since the geometry of a thin plate element is the same as that of a plane stress element. Under the assumption of the classical thin plate bending theory, the governing equation of wave motion of an elastic, isotropic thin plate with constant thickness is written as

$$D_f \left(\frac{\partial^4 W}{\partial x^4} + 2 \frac{\partial^4 W}{\partial x^2 \partial y^2} + \frac{\partial^4 W}{\partial y^4} \right) + \rho \frac{\partial^2 W}{\partial t^2} = q(x, y, t), \quad (1)$$

where W is the transverse displacement of the middle plane of the thin plate; ρ is the mass per unit area of the thin plate; q is the dynamic load carried by per unit area of the thin plate and D_f is the flexural rigidity of the thin plate and can be expressed in this particular case as

$$D_f = \frac{ET^3}{12(1 - \mu^2)}, \quad (2)$$

where E and μ are the elastic modulus and Poisson's ratio of the plate material; T is the thickness of the thin plate.

It is noted that eqn (1) is not a standard wave equation since the quantity, $\sqrt{D_f/\rho}$, does not have the dimension of velocity. For a thin plate of infinite size, it has been demonstrated that the thin plate, from the wave motion point of view, is a dispersive system because the wave within the plate travels at the phase velocity which depends on both the exciting frequency and the material property of the plate, instead of at the physical velocity which depends only on the material property of the plate. Owing to this reason, it is very difficult and complicated to obtain analytical solutions to the governing equation of wave motion of a thin plate, so that it becomes a common practice to get the equation solved by the numerical method. If the external force is ignored and the finite element method is used, eqn (1) can be discretized into a finite element equation under free-vibration conditions as follows:

$$(-\omega^2[M] + [K])\{\Delta\} = \{0\}, \quad (3)$$

where ω is the natural frequency of a system, $\{\Delta\}$ is the corresponding natural mode of the system, $[M]$ and $[K]$ are the global mass matrix and stiffness matrix of the system, and they are obtained by assembling the mass matrices and stiffness matrices of relevant finite elements in the system.

As was mentioned in the introduction, the main purpose of this study is to develop an error estimator/corrector to correct the predicted natural frequencies of a thin plate vibration problem in the finite element analysis. To accomplish this purpose, explicit expressions for the mass matrix and stiffness matrix of a thin plate bending element are needed in the analysis. Although some explicit expressions are available, they are briefly derived in this section due to the following two reasons. One is that the mass matrix and stiffness matrix of a thin plate bending element are basic formulae and therefore need to be included in this paper for the sake of completeness. The other is that the existing explicit expressions for the property matrices of a thin plate were derived either in a left-hand coordinate system or in a right-hand coordinate system, but the positive direction of rotation, θ_y , and its corresponding linear moment was opposite to the positive direction of the y axis [3, 4], which are not preferred by a potential practitioner of the finite element method.

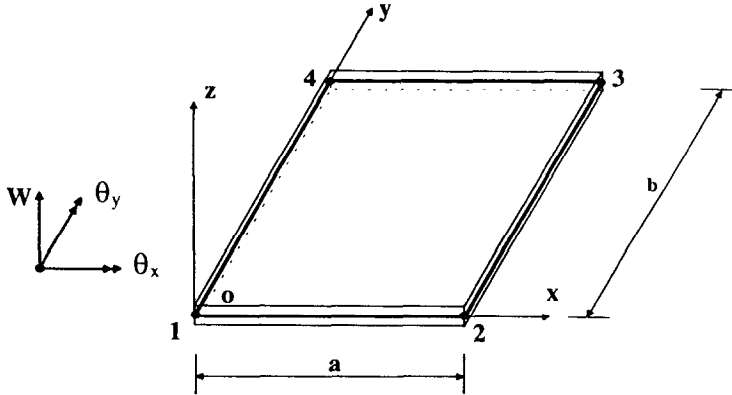


Fig. 1. A four-node thin plate bending element.

For a four-node thin plate element shown in Fig. 1, the displacement field of the element was defined as [5]

$$W = \{p\}^T \{a\}, \quad (4)$$

where

$$\begin{aligned} \{p\}^T &= \{1 \quad x \quad y \quad x^2 \quad xy \quad y^2 \quad x^3 \quad x^2y \quad xy^2 \quad y^3 \quad x^3y \quad xy^3\} \\ \{a\}^T &= \{a_1 \quad a_2 \quad a_3 \quad a_4 \quad a_5 \quad a_6 \quad a_7 \quad a_8 \quad a_9 \quad a_{10} \quad a_{11} \quad a_{12}\}. \end{aligned} \quad (5)$$

Using the definition in Fig. 1, there exist the following relationships:

$$\theta_x = \frac{\partial W}{\partial y}, \quad \theta_y = -\frac{\partial W}{\partial x}. \quad (6)$$

It should be pointed out that for θ_y , the negative value of the derivative is due to the fact that a positive rotation about the y axis causes a negative value of the transverse displacement, W .

In the finite element analysis, $\{a\}$ can be determined by considering the nodal displacements and rotations of the element.

$$\{\Delta\}^e = [C]\{a\}, \quad (7)$$

where $[C]$ is a coefficient matrix and $\{\Delta\}^e$ is a vector of the nodal displacements and rotations of the element.

$$\{\Delta\}^e = \{W_1 \quad \theta_{x1} \quad \theta_{y1} \quad W_2 \quad \theta_{x2} \quad \theta_{y2} \quad W_3 \quad \theta_{x3} \quad \theta_{y3} \quad W_4 \quad \theta_{x4} \quad \theta_{y4}\}^T \quad (8)$$

$$[C] = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 1 & a & 0 & a^2 & 0 & 0 & a^3 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & a & 0 & 0 & a^2 & 0 & 0 & a^3 & 0 \\ 0 & -1 & 0 & -2a & 0 & 0 & -3a^2 & 0 & 0 & 0 & 0 & 0 \\ 1 & a & b & a^2 & ab & b^2 & a^3 & a^2b & ab^2 & b^3 & a^3b & ab^3 \\ 0 & 0 & 1 & 0 & a & 2b & 0 & a^2 & 2ab & 3b^2 & a^3 & 3ab^2 \\ 0 & -1 & 0 & -2a & -b & 0 & -3a^2 & -2ab & -b^2 & 0 & -3a^2b & -b^3 \\ 1 & 0 & b & 0 & 0 & b^2 & 0 & 0 & 0 & b^3 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 2b & 0 & 0 & 0 & 3b^2 & 0 & 0 \\ 0 & -1 & 0 & 0 & -b & 0 & 0 & 0 & -b^2 & 0 & 0 & -b^3 \end{bmatrix}. \quad (9)$$

Substituting eqn (7) into eqn (4), the following equation yields:

$$[N] = \{p\}^T [C]^{-1}, \tag{10}$$

where $[N]$ is the shape function matrix of the element and can be further defined as

$$[N] = [N_1 \ N_2 \ N_3 \ N_4 \ N_5 \ N_6 \ N_7 \ N_8 \ N_9 \ N_{10} \ N_{11} \ N_{12}]. \tag{11}$$

Note that the inverse of matrix $[C]$ is derived as

$$[C]^{-1} = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{3}{a^2} & 0 & \frac{2}{a} & \frac{3}{a^2} & 0 & \frac{1}{a} & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{1}{ab} & -\frac{1}{a} & \frac{1}{b} & \frac{1}{ab} & \frac{1}{a} & 0 & -\frac{1}{ab} & 0 & 0 & \frac{1}{ab} & 0 & -\frac{1}{b} \\ -\frac{3}{b^2} & -\frac{2}{b} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3}{b^2} & -\frac{1}{b} & 0 \\ \frac{2}{a^3} & 0 & -\frac{1}{a^2} & -\frac{2}{a^3} & 0 & -\frac{1}{a^2} & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{3}{a^2b} & 0 & -\frac{2}{ab} & -\frac{3}{a^2b} & 0 & -\frac{1}{ab} & \frac{3}{a^2b} & 0 & \frac{1}{ab} & -\frac{3}{a^2b} & 0 & \frac{2}{ab} \\ \frac{3}{ab^2} & \frac{2}{ab} & 0 & -\frac{3}{ab^2} & -\frac{2}{ab} & 0 & \frac{3}{ab^2} & -\frac{1}{ab} & 0 & -\frac{3}{ab^2} & \frac{1}{ab} & 0 \\ \frac{2}{b^3} & \frac{1}{b^2} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{2}{b^3} & \frac{1}{b^2} & 0 \\ -\frac{2}{a^3b} & 0 & \frac{1}{a^2b} & \frac{2}{a^3b} & 0 & \frac{1}{a^2b} & -\frac{2}{a^3b} & 0 & -\frac{1}{a^2b} & \frac{2}{a^3b} & 0 & -\frac{1}{a^2b} \\ -\frac{2}{ab^3} & -\frac{1}{ab^2} & 0 & \frac{2}{ab^3} & \frac{1}{ab^2} & 0 & -\frac{2}{ab^3} & \frac{1}{ab^2} & 0 & \frac{2}{ab^3} & -\frac{1}{ab^2} & 0 \end{bmatrix}. \tag{12}$$

Substituting eqn (12) into eqn (10), the elements in the shape function matrix, $[N]$, can be derived as follows:

$$\begin{aligned} N_1 &= (1 - 3\xi^2 + 2\xi^3 - 3\eta^2 + 2\eta^3) - \xi\eta(1 - 3\xi - 3\eta + 2\xi^2 + 2\eta^2) \\ N_2 &= y(1 - \xi)(1 - \eta)^2 \\ N_3 &= -x(1 - \xi)^2(1 - \eta) \end{aligned} \tag{13a}$$

$$\begin{aligned} N_4 &= \xi(1 - \eta)(3\xi - 2\xi^2 + \eta - 2\eta^2) \\ N_5 &= y\xi(1 - \eta)^2 \\ N_6 &= x(\xi - \xi^2)(1 - \eta) \end{aligned} \tag{13b}$$

$$\begin{aligned} N_7 &= \xi\eta(-1 + 3\xi + 3\eta - 2\xi^2 - 2\eta^2) \\ N_8 &= y\xi(\eta^2 - \eta) \\ N_9 &= x\eta(\xi - \xi^2) \end{aligned} \tag{13c}$$

$$N_{10} = (1 - \xi)\eta(3\eta - 2\eta^2 + \xi - 2\xi^2)$$

$$N_{11} = y(1 - \xi)(\eta^2 - \eta)$$

$$N_{12} = -x\eta(1 - \xi)^2, \quad (13d)$$

where

$$\xi = \frac{x}{a}, \quad \eta = \frac{y}{b}. \quad (14)$$

Like the strain vector of a plane stress element in two-dimensional elasticity, the curvature vector of the thin plate element is expressed as

$$\begin{Bmatrix} \kappa_x \\ \kappa_y \\ \kappa_{xy} \end{Bmatrix} = \begin{Bmatrix} -\frac{\partial^2 W}{\partial x^2} \\ -\frac{\partial^2 W}{\partial y^2} \\ -2\frac{\partial^2 W}{\partial x \partial y} \end{Bmatrix} = [B]\{\Delta\}^e, \quad (15)$$

where $[B]$ is the strain matrix of the thin plate element and can be further defined in partition form as

$$[B] = [[B_1] \quad [B_2] \quad [B_3] \quad [B_4]], \quad (16)$$

where

$$[B_i] = \begin{bmatrix} -\frac{\partial^2 N_{3i-2}}{\partial x^2} & -\frac{\partial^2 N_{3i-1}}{\partial x^2} & -\frac{\partial^2 N_{3i}}{\partial x^2} \\ -\frac{\partial^2 N_{3i-2}}{\partial y^2} & -\frac{\partial^2 N_{3i-1}}{\partial y^2} & -\frac{\partial^2 N_{3i}}{\partial y^2} \\ -2\frac{\partial^2 N_{3i-2}}{\partial x \partial y} & -2\frac{\partial^2 N_{3i-1}}{\partial x \partial y} & -2\frac{\partial^2 N_{3i}}{\partial x \partial y} \end{bmatrix} \quad (i = 1, 2, 3, 4). \quad (17)$$

Substituting eqn (13a-d) into eqn (17), the submatrices in eqn (16) are expressed as

$$[B_1] = \begin{bmatrix} \frac{6}{a^2}(1 - 2\xi)(1 - \eta) & 0 & -\frac{2}{a}(2 - 3\xi)(1 - \eta) \\ \frac{6}{b^2}(1 - \xi)(1 - 2\eta) & \frac{2}{b}(1 - \xi)(2 - 3\eta) & 0 \\ \frac{2}{ab}[1 - 6(\xi + \eta) + 6(\xi^2 + \eta^2)] & \frac{2}{a}(1 - \eta)(1 - 3\eta) & -\frac{2}{b}(1 - \xi)(1 - 3\xi) \end{bmatrix} \quad (18a)$$

$$[B_2] = \begin{bmatrix} -\frac{6}{a^2}(1 - 2\xi)(1 - \eta) & 0 & \frac{-2}{a}(1 - 3\xi)(1 - \eta) \\ \frac{6}{b^2}\xi(1 - 2\eta) & \frac{2}{b}\xi(2 - 3\eta) & 0 \\ \frac{-2}{ab}[1 - 6(\xi + \eta) + 6(\xi^2 + \eta^2)] & \frac{-2}{a}(1 - \eta)(1 - 3\eta) & \frac{2}{b}(2\xi - 3\xi^2) \end{bmatrix} \quad (18b)$$

$$[B_3] = \begin{bmatrix} -\frac{6}{a^2}(1-2\xi)\eta & 0 & -\frac{2}{a}(1-3\xi)\eta \\ -\frac{6}{b^2}\xi(1-2\eta) & \frac{2}{b}\xi(1-3\eta) & 0 \\ \frac{2}{ab}[1-6(\xi+\eta)+6(\xi^2+\eta^2)] & \frac{2}{a}(2\eta-3\eta^2) & -\frac{2}{b}(2\xi-3\xi^2) \end{bmatrix} \quad (18c)$$

$$[B_4] = \begin{bmatrix} \frac{6}{a^2}(1-2\xi)\eta & 0 & -\frac{2}{a}(2-3\xi)\eta \\ -\frac{6}{b^2}(1-\xi)(1-2\eta) & \frac{2}{b}(1-\xi)(1-3\eta) & 0 \\ -\frac{2}{ab}[1-6(\xi+\eta)+6(\xi^2+\eta^2)] & -\frac{2}{a}(2\eta-3\eta^2) & \frac{2}{b}(1-\xi)(1-3\xi) \end{bmatrix}. \quad (18d)$$

Similar to a plane stress element in two-dimensional elasticity, the stiffness matrix of the thin plate bending element is expressed as

$$[K]^e = \int_0^a \int_0^b [B]^T [D] [B] \, dx \, dy, \quad (19)$$

where $[D]$ is the constitutive matrix of the element. In the case of an elastic, isotropic thin plate, $[D]$ is written as

$$[D] = D_f \begin{bmatrix} 1 & \mu & 0 \\ \mu & 1 & 0 \\ 0 & 0 & \frac{1-\mu}{2} \end{bmatrix}. \quad (20)$$

Thus, the element of the stiffness matrix, $[K]^e$, is expressed as

$$K_{ij} = (ab) \int_0^1 \int_0^1 [D_{11}(B_{1i}B_{1j} + B_{2i}B_{2j}) + D_{12}(B_{2i}B_{1j} + B_{1i}B_{2j}) + D_{33}B_{3i}B_{3j}] \, d\xi \, d\eta \quad (i = 1, 2, \dots, 12, j = 1, 2, \dots, 12). \quad (21)$$

For the ease of expression, the stiffness matrix of the thin plate element can be partitioned as follows:

$$[K]^e = \begin{bmatrix} [K_{11}] & [K_{21}]^T & [K_{31}]^T & [K_{41}]^T \\ [K_{21}] & [K_{22}] & [K_{32}]^T & [K_{42}]^T \\ [K_{31}] & [K_{32}] & [K_{33}] & [K_{43}]^T \\ [K_{41}] & [K_{42}] & [K_{43}] & [K_{44}] \end{bmatrix}, \quad (22)$$

where

$$[K_{11}] = ab \begin{bmatrix} \left(\frac{4}{a^4} + \frac{4}{b^4}\right)D_{11} + \frac{2}{a^2b^2}D_{12} + \frac{28}{5a^2b^2}D_{33} & \frac{2}{b^3}D_{11} + \frac{1}{a^2b}D_{12} + \frac{2}{5a^2b}D_{33} & -\frac{2}{a^3}D_{11} - \frac{1}{ab^2}D_{12} - \frac{2}{5ab^2}D_{33} \\ \frac{2}{b^3}D_{11} + \frac{1}{a^2b}D_{12} + \frac{2}{5a^2b}D_{33} & \frac{4}{3b^2}D_{11} + \frac{8}{15a^2}D_{33} & -\frac{1}{ab}D_{12} \\ -\frac{2}{a^3}D_{11} - \frac{1}{ab^2}D_{12} - \frac{2}{5ab^2}D_{33} & -\frac{1}{ab}D_{12} & \frac{4}{3a^2}D_{11} + \frac{8}{15b^2}D_{33} \end{bmatrix}$$

$$[K_{21}] =$$

$$ab \begin{bmatrix} \left(\frac{-4}{a^4} + \frac{2}{b^4} \right) D_{11} - \frac{2}{a^2 b^2} D_{12} - \frac{28}{5a^2 b^2} D_{33} & \frac{1}{b^3} D_{11} - \frac{1}{a^2 b} D_{12} - \frac{2}{5a^2 b} D_{33} & \frac{2}{a^3} D_{11} + \frac{2}{5ab^2} D_{33} \\ \frac{1}{b^3} D_{11} - \frac{1}{a^2 b} D_{12} - \frac{2}{5a^2 b} D_{33} & \frac{2}{3b^2} D_{11} - \frac{8}{15a^2} D_{33} & 0 \\ -\frac{2}{a^3} D_{11} - \frac{2}{5ab^2} D_{33} & 0 & \frac{2}{3a^2} D_{11} - \frac{2}{15b^2} D_{33} \end{bmatrix}$$

$$[K_{31}] =$$

$$ab \begin{bmatrix} \left(\frac{-2}{a^4} - \frac{2}{b^4} \right) D_{11} + \frac{2}{a^2 b^2} D_{12} + \frac{28}{5a^2 b^2} D_{33} & \frac{-1}{b^3} D_{11} + \frac{2}{5a^2 b} D_{33} & \frac{1}{a^3} D_{11} - \frac{2}{5ab^2} D_{33} \\ \frac{1}{b^3} D_{11} - \frac{2}{5a^2 b} D_{33} & \frac{1}{3b^2} D_{11} + \frac{2}{15a^2} D_{33} & 0 \\ \frac{-1}{a^3} D_{11} + \frac{2}{5ab^2} D_{33} & 0 & \frac{1}{3a^2} D_{11} + \frac{2}{15b^2} D_{33} \end{bmatrix}$$

$$[K_{41}] =$$

$$ab \begin{bmatrix} \left(\frac{2}{a^4} - \frac{4}{b^4} \right) D_{11} - \frac{2}{a^2 b^2} D_{12} - \frac{28}{5a^2 b^2} D_{33} & \frac{-2}{b^3} D_{11} - \frac{2}{5a^2 b} D_{33} & \frac{-1}{a^3} D_{11} + \frac{1}{ab^2} D_{12} + \frac{2}{5ab^2} D_{33} \\ \frac{2}{b^3} D_{11} + \frac{2}{5a^2 b} D_{33} & \frac{2}{3b^2} D_{11} - \frac{2}{15a^2} D_{33} & 0 \\ \frac{-1}{a^3} D_{11} + \frac{1}{ab^2} D_{12} + \frac{2}{5ab^2} D_{33} & 0 & \frac{2}{3a^2} D_{11} - \frac{8}{15b^2} D_{33} \end{bmatrix}$$

$$[K_{22}] =$$

$$ab \begin{bmatrix} \left(\frac{4}{a^4} + \frac{4}{b^4} \right) D_{11} + \frac{2}{a^2 b^2} D_{12} + \frac{28}{5a^2 b^2} D_{33} & \frac{2}{b^3} D_{11} + \frac{1}{a^2 b} D_{12} + \frac{2}{5a^2 b} D_{33} & \frac{2}{a^3} D_{11} + \frac{1}{ab^2} D_{12} + \frac{2}{5ab^2} D_{33} \\ \frac{2}{b^3} D_{11} + \frac{1}{a^2 b} D_{12} + \frac{2}{5a^2 b} D_{33} & \frac{4}{3b^2} D_{11} + \frac{8}{15a^2} D_{33} & \frac{1}{ab} D_{12} \\ \frac{2}{a^3} D_{11} + \frac{1}{ab^2} D_{12} + \frac{2}{5ab^2} D_{33} & \frac{1}{ab} D_{12} & \frac{4}{3a^2} D_{11} + \frac{8}{15b^2} D_{33} \end{bmatrix}$$

$$[K_{32}] =$$

$$ab \begin{bmatrix} \left(\frac{2}{a^4} - \frac{4}{b^4} \right) D_{11} - \frac{2}{a^2 b^2} D_{12} - \frac{28}{5a^2 b^2} D_{33} & \frac{-2}{b^3} D_{11} - \frac{2}{5a^2 b} D_{33} & \frac{1}{a^3} D_{11} - \frac{1}{ab^2} D_{12} - \frac{2}{5ab^2} D_{33} \\ \frac{2}{b^3} D_{11} + \frac{2}{5a^2 b} D_{33} & \frac{2}{3b^2} D_{11} - \frac{2}{15a^2} D_{33} & 0 \\ \frac{1}{a^3} D_{11} - \frac{1}{ab^2} D_{12} - \frac{2}{5ab^2} D_{33} & 0 & \frac{2}{3a^2} D_{11} - \frac{8}{15b^2} D_{33} \end{bmatrix}$$

$$[K_{42}] =$$

$$ab \begin{bmatrix} \left(\frac{-2}{a^4} - \frac{2}{b^4} \right) D_{11} + \frac{2}{a^2 b^2} D_{12} + \frac{28}{5a^2 b^2} D_{33} & \frac{-1}{b^3} D_{11} + \frac{2}{5a^2 b} D_{33} & \frac{-1}{a^3} D_{11} + \frac{2}{5ab^2} D_{33} \\ \frac{1}{b^3} D_{11} - \frac{2}{5a^2 b} D_{33} & \frac{1}{3b^2} D_{11} + \frac{2}{15a^2} D_{33} & 0 \\ \frac{1}{a^3} D_{11} - \frac{2}{5ab^2} D_{33} & 0 & \frac{1}{3a^2} D_{11} + \frac{2}{15b^2} D_{33} \end{bmatrix}$$

$$[K_{33}] =$$

$$ab \begin{bmatrix} \left(\frac{4}{a^4} + \frac{4}{b^4} \right) D_{11} + \frac{2}{a^2 b^2} D_{12} + \frac{28}{5a^2 b^2} D_{33} & \frac{-2}{b^3} D_{11} - \frac{1}{a^2 b} D_{12} - \frac{2}{5a^2 b} D_{33} & \frac{2}{a^3} D_{11} + \frac{1}{ab^2} D_{12} + \frac{2}{5ab^2} D_{33} \\ \frac{-2}{b^3} D_{11} - \frac{1}{a^2 b} D_{12} - \frac{2}{5a^2 b} D_{33} & \frac{4}{3b^2} D_{11} + \frac{8}{15a^2} D_{33} & -\frac{1}{ab} D_{12} \\ \frac{2}{a^3} D_{11} + \frac{1}{ab^2} D_{12} + \frac{2}{5ab^2} D_{33} & -\frac{1}{ab} D_{12} & \frac{4}{3a^2} D_{11} + \frac{8}{15b^2} D_{33} \end{bmatrix}$$

$$[K_{43}] =$$

$$ab \begin{bmatrix} \left(\frac{-4}{a^4} + \frac{2}{b^4} \right) D_{11} - \frac{2}{a^2 b^2} D_{12} - \frac{28}{5a^2 b^2} D_{33} & \frac{-1}{b^3} D_{11} + \frac{1}{a^2 b} D_{12} + \frac{2}{5a^2 b} D_{33} & \frac{-2}{a^3} D_{11} - \frac{2}{5ab^2} D_{33} \\ \frac{-1}{b^3} D_{11} + \frac{1}{a^2 b} D_{12} + \frac{2}{5a^2 b} D_{33} & \frac{2}{3b^2} D_{11} - \frac{8}{15a^2} D_{33} & 0 \\ \frac{2}{a^3} D_{11} + \frac{2}{5ab^2} D_{33} & 0 & \frac{2}{3a^2} D_{11} - \frac{2}{15b^2} D_{33} \end{bmatrix}$$

$$[K_{44}] =$$

$$ab \begin{bmatrix} \left(\frac{4}{a^4} + \frac{4}{b^4} \right) D_{11} + \frac{2}{a^2 b^2} D_{12} + \frac{28}{5a^2 b^2} D_{33} & \frac{-2}{b^3} D_{11} - \frac{1}{a^2 b} D_{12} - \frac{2}{5a^2 b} D_{33} & \frac{-2}{a^3} D_{11} - \frac{1}{ab^2} D_{12} - \frac{2}{5ab^2} D_{33} \\ \frac{-2}{b^3} D_{11} - \frac{1}{a^2 b} D_{12} - \frac{2}{5a^2 b} D_{33} & \frac{4}{3b^2} D_{11} + \frac{8}{15a^2} D_{33} & \frac{1}{ab} D_{12} \\ \frac{-2}{a^3} D_{11} - \frac{1}{ab^2} D_{12} - \frac{2}{5ab^2} D_{33} & \frac{1}{ab} D_{12} & \frac{4}{3a^2} D_{11} + \frac{8}{15b^2} D_{33} \end{bmatrix}$$

(23)

Similarly, the consistent mass matrix of the thin plate bending element can be derived from the displacement shape functions of the element as follows:

$$[M]^e = \int_0^a \int_0^b [N]^T \rho [N] dx dy. \quad (24)$$

This matrix is rewritten in partition form as

$$[M]^e = \begin{bmatrix} [M_{11}] & [M_{21}]^T \\ [M_{21}] & [M_{22}] \end{bmatrix}, \quad (25)$$

where

$$\begin{aligned}
 [M_{11}] &= \frac{\rho ab}{25,200} \begin{bmatrix} 3454 & 461b & -461b & 1226 & 199b & 274a \\ 461b & 80b^2 & -63ab & 199b & 40b^2 & 42ab \\ -461a & -63ab & 80a^2 & -274a & -42ab & -60a^2 \\ 1226 & 199b & -274a & 3454 & 461b & 461a \\ 199b & 40b^2 & -42ab & 461b & 80b^2 & 63ab \\ 274a & 42ab & -60a^2 & 461a & 63ab & 80a^2 \end{bmatrix} \\
 [M_{21}] &= \frac{\rho ab}{25,200} \begin{bmatrix} 394 & 116b & -116a & 1226 & 274b & 199a \\ -116b & -30b^2 & 28ab & -274b & -60b^2 & -42ab \\ 116a & 28ab & -30a^2 & 199a & 42ab & 40a^2 \\ 1226 & 274b & -199a & 394 & 116b & 116a \\ -274b & -60b^2 & 42ab & -116b & -30b^2 & -28ab \\ -199a & -42ab & 40a^2 & -116a & -28ab & -30a^2 \end{bmatrix} \\
 [M_{22}] &= \frac{\rho ab}{25,200} \begin{bmatrix} 3454 & -461b & 461a & 1226 & -199b & -274a \\ -461b & 80b^2 & -63ab & -199b & 40b^2 & 42ab \\ 461a & -63ab & 80a^2 & 274a & -42ab & -60a^2 \\ 1226 & -199b & 274a & 3454 & -461b & -461a \\ -199b & 40b^2 & -42ab & -461b & 80b^2 & 63ab \\ -274a & 42ab & -60a^2 & -461a & 63ab & 80a^2 \end{bmatrix}. \tag{26}
 \end{aligned}$$

3. ASYMPTOTIC FORMULA FOR NATURAL FREQUENCIES OF THIN PLATES

In the limit, when the element size approaches zero, the discretized finite element system approaches a continuous system. As a result, the approximate solution from the finite element analysis approaches an exact solution. Although this idea has been successfully applied to the error estimate/correction of natural frequencies of one-dimensional bars and beams [12, 13], as well as two-dimensional membranes [14] very recently, it has not been used for the error estimate/correction of natural frequencies of thin plates so far. In this section, the above-mentioned idea will be used to derive an asymptotic formula for the predicted natural frequency of a thin plate in the finite element analysis.

As shown in Fig. 2, a typical node (node 5) in the finite element system is considered. Since node 5 is surrounded by four thin plate elements, it is sufficient to express the equation of motion for this typical node

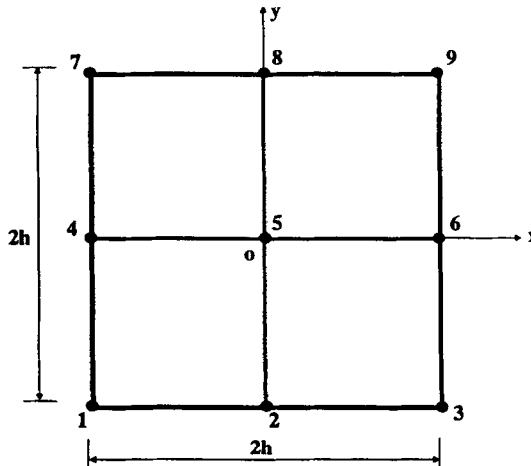


Fig. 2. A typical node connection by four elements.

in terms of the nine nodes in the figure. Letting a and b equal h , the stiffness matrix of a square thin plate bending element used in Fig. 2 can be derived from eqn (22) as

$$[K]_{sq}^e = \begin{bmatrix} [K_{11}^*] & [K_{21}^*]^T \\ [K_{21}^*] & [K_{22}^*] \end{bmatrix}, \quad (27)$$

where

$$[K_{11}^*] = \frac{D_f}{15h^2} \begin{bmatrix} 162 - 12\mu & h(33 + 12\mu) & h(-33 - 12\mu) & -72 + 12\mu & h(12 - 12\mu) & h(-33 + 3\mu) \\ h(33 + 12\mu) & h^2(24 - 4\mu) & h^2(-15\mu) & h(12 - 12\mu) & h^2(6 + 4\mu) & 0 \\ h(-33 - 12\mu) & h^2(-15\mu) & h^2(24 - 4\mu) & h(33 - 3\mu) & 0 & h^2(9 + \mu) \\ -72 + 12\mu & h(12 - 12\mu) & h(33 - 3\mu) & 162 - 12\mu & h(33 + 12\mu) & h(33 + 12\mu) \\ h(12 - 12\mu) & h^2(6 + 4\mu) & 0 & h(33 + 12\mu) & h^2(24 - 4\mu) & h^2(15\mu) \\ h(-33 + 3\mu) & 0 & h^2(9 + \mu) & h(33 + 12\mu) & h^2(15\mu) & h^2(24 - 4\mu) \end{bmatrix}$$

$$[K_{21}^*] = \frac{D_f}{15h^2} \begin{bmatrix} -18 - 12\mu & h(-12 - 3\mu) & h(12 + 3\mu) & -72 + 12\mu & h(-33 + 3\mu) & h(12 - 12\mu) \\ h(12 + 3\mu) & h^2(6 - \mu) & 0 & h(33 - 3\mu) & h^2(9 + \mu) & 0 \\ h(-12 - 3\mu) & 0 & h^2(6 - \mu) & h(12 - 12\mu) & 0 & h^2(6 + 4\mu) \\ -72 + 12\mu & h(-33 + 3\mu) & h(-12 + 12\mu) & -18 - 12\mu & h(-12 - 3\mu) & h(-12 - 3\mu) \\ h(33 - 3\mu) & h^2(9 + \mu) & 0 & h(12 + 3\mu) & h^2(6 - \mu) & 0 \\ h(-12 + 12\mu) & 0 & h^2(6 + 4\mu) & h(12 + 3\mu) & 0 & h^2(6 - \mu) \end{bmatrix}$$

$$[K_{22}^*] = \frac{D_f}{15h^2} \begin{bmatrix} 162 - 12\mu & h(-33 - 12\mu) & h(33 + 12\mu) & -72 + 12\mu & h(-12 + 12\mu) & h(33 - 3\mu) \\ h(-33 - 12\mu) & h^2(24 - 4\mu) & h^2(-15\mu) & h(-12 + 12\mu) & h^2(6 + 4\mu) & 0 \\ h(33 + 12\mu) & h^2(-15\mu) & h^2(24 - 4\mu) & h(-33 + 3\mu) & 0 & h^2(9 + \mu) \\ -72 + 12\mu & h(-12 + 12\mu) & h(-33 + 3\mu) & 162 - 12\mu & h(-33 - 12\mu) & h(-33 - 12\mu) \\ h(-12 + 12\mu) & h^2(6 + 4\mu) & 0 & h(-33 - 12\mu) & h^2(24 - 4\mu) & h^2(15\mu) \\ h(33 - 3\mu) & 0 & h^2(9 + \mu) & h(-33 - 12\mu) & h^2(15\mu) & h^2(24 - 4\mu) \end{bmatrix}. \quad (28)$$

In a similar way, the consistent mass matrix of the square thin plate bending element used in Fig. 2 is derived from eqn (25) as

$$[M]_{sq}^e = \begin{bmatrix} [M_{11}^*] & [M_{21}^*]^T \\ [M_{21}^*] & [M_{22}^*] \end{bmatrix}, \quad (29)$$

where

$$[M_{11}^*] = \frac{\rho h^2}{25,200} \begin{bmatrix} 3454 & 461h & -461h & 1226 & 199h & 274h \\ 461h & 80h^2 & -63h^2 & 199h & 40h^2 & 42h^2 \\ -461h & -63h^2 & 80h^2 & -274h & -42h^2 & -60h^2 \\ 1226 & 199h & -274h & 3454 & 461h & 461h \\ 199h & 40h^2 & -42h^2 & 461h & 80h^2 & 63h^2 \\ 274h & 42h^2 & -60h^2 & 461h & 63h^2 & 80h^2 \end{bmatrix}$$

$$[M_{21}^*] = \frac{\rho h^2}{25,200} \begin{bmatrix} 394 & 116h & -116h & 1226 & 274h & 199h \\ -116h & -30h^2 & 28h^2 & -274h & -60h^2 & -42h^2 \\ 116h & 28h^2 & -30h^2 & 199h & 42h^2 & 40h^2 \\ 1226 & 274h & -199h & 394 & 116h & 116h \\ -274h & -60h^2 & 42h^2 & -116h & -30h^2 & -28h^2 \\ -199h & -42h^2 & 40h^2 & -116h & -28h^2 & -30h^2 \end{bmatrix}$$

$$[M_{22}^*] = \frac{\rho h^2}{25,200} \begin{bmatrix} 3454 & -461h & 461h & 1226 & -199h & -274h \\ -461h & 80h^2 & -63h^2 & -199h & 40h^2 & 42h^2 \\ 461h & -63h^2 & 80h^2 & 274h & -42h^2 & -60h^2 \\ 1226 & -199h & 274h & 3454 & -461h & -461h \\ -199h & 40h^2 & -42h^2 & -461h & 80h^2 & 63h^2 \\ -274h & 42h^2 & -60h^2 & -461h & 63h^2 & 80h^2 \end{bmatrix}. \quad (30)$$

Through assembling the stiffness matrices and mass matrices of the four thin plate bending elements in Fig. 2, the finite element equations of node 5 are expressed as

$$\begin{aligned} & -\beta[394(W_1 + W_3 + W_7 + W_9) + 2452(W_2 + W_4 + W_6 + W_8) + 13816W_5 \\ & + 116h(\theta_{x1} + \theta_{x3} - \theta_{x7} - \theta_{x9}) + 548h(\theta_{x2} - \theta_{x8}) \\ & + 116h(\theta_{y3} + \theta_{y9} - \theta_{y1} - \theta_{y7}) + 548h(\theta_{y6} - \theta_{y4})] \\ & + [-(18 + 12\mu)(W_1 + W_3 + W_7 + W_9) - (144 - 24\mu)(W_2 + W_4 + W_6 + W_8) + (648 - 48\mu)W_5 \\ & - h(12 + 3\mu)(\theta_{x1} + \theta_{x3} - \theta_{x7} - \theta_{x9}) - h(66 - 6\mu)(\theta_{x2} - \theta_{x8}) \\ & - h(12 + 3\mu)(\theta_{y3} + \theta_{y9} - \theta_{y1} - \theta_{y7}) - h(66 - 6\mu)(\theta_{y6} - \theta_{y4})] = 0 \end{aligned} \quad (31a)$$

$$\begin{aligned} & -\beta[-30h^2(\theta_{x1} + \theta_{x3} + \theta_{x7} + \theta_{x9}) - 120h^2(\theta_{x2} + \theta_{x8}) + 80h^2(\theta_{x4} + \theta_{x6}) + 320h^2\theta_{x5} \\ & + 116h(W_7 + W_9 - W_1 - W_3) + 548h(W_8 - W_2) + 28h^2(\theta_{y1} - \theta_{y3} - \theta_{y7} + \theta_{y9})] \\ & + [h^2(6 - \mu)(\theta_{x1} + \theta_{x3} + \theta_{x7} + \theta_{x9}) + h^2(18 + 2\mu)(\theta_{x2} + \theta_{x8}) + h^2(12 + 8\mu)(\theta_{x4} + \theta_{x6}) \\ & + h^2(96 - 16\mu)\theta_{x5} - h(12 + 3\mu)(W_7 + W_9 - W_1 - W_3) + h(66 - 6\mu)(W_2 - W_8)] = 0 \end{aligned} \quad (31b)$$

$$\begin{aligned} & -\beta[-30h^2(\theta_{y1} + \theta_{y3} + \theta_{y7} + \theta_{y9}) + 80h^2(\theta_{y2} + \theta_{y8}) - 120h^2(\theta_{y4} + \theta_{y6}) + 320h^2\theta_{y5} \\ & + 116h(W_1 - W_3 + W_7 - W_9) + 548h(W_4 - W_6) + 28h^2(\theta_{x1} - \theta_{x3} - \theta_{x7} + \theta_{x9})] \\ & + [h^2(6 - \mu)(\theta_{y1} + \theta_{y3} + \theta_{y7} + \theta_{y9}) + h^2(18 + 2\mu)(\theta_{y4} + \theta_{y6}) + h^2(12 + 8\mu)(\theta_{y2} + \theta_{y8}) \\ & + h^2(96 - 16\mu)\theta_{y5} + h(12 + 3\mu)(W_3 - W_1 + W_9 - W_7) + h(66 - 6\mu)(W_6 - W_4)] = 0, \end{aligned} \quad (31c)$$

where

$$\beta = \frac{\omega^2 \rho h^4}{1680 D_f}. \quad (32)$$

For a vibration or wave propagation problem in a thin plate under harmonic loading, the general form of its transverse displacement and rotations are expressed as

$$\begin{aligned} W &= C_1 \exp[i(k_x x + k_y y)] \\ \theta_x &= C_2 \exp[i(k_x x + k_y y)] \\ \theta_y &= C_3 \exp[i(k_x x + k_y y)], \end{aligned} \quad (33)$$

where C_1 , C_2 and C_3 are constants, k_x and k_y are wave numbers of the x - and y -direction, respectively.

Substituting eqn (33) into eqn (31a-c) yields the following equations:

$$\begin{aligned} A_{11} C_1 + A_{12} C_2 + A_{13} C_3 &= 0 \\ A_{21} C_1 + A_{22} C_2 + A_{23} C_3 &= 0 \\ A_{31} C_1 + A_{32} C_2 + A_{33} C_3 &= 0, \end{aligned} \quad (34)$$

where

$$\begin{aligned}
 A_{11} &= [-394\beta - (18 + 12\mu)] \{2 \cos[(k_x + k_y)h] + 2 \cos[(k_x - k_y)h]\} \\
 &\quad - [2452\beta + (144 - 24\mu)] [2 \cos(k_x h) + 2 \cos(k_y h)] - (13816\beta - 648 + 48\mu) \\
 A_{12} &= [116h\beta + h(12 + 3\mu)] \{2i \sin[(k_x + k_y)h] - 2i \sin[(k_x - k_y)h]\} + [548h\beta + h(66 - 6\mu)] 2i \sin(k_y h) \\
 A_{13} &= -[116h\beta + h(12 + 3\mu)] \{2i \sin[(k_x + k_y)h] + 2i \sin[(k_x - k_y)h]\} - [548h\beta + h(66 - 6\mu)] 2i \sin(k_x h) \\
 A_{21} &= -[116h\beta + h(12 + 3\mu)] \{2i \sin[(k_x + k_y)h] - 2i \sin[(k_x - k_y)h]\} - [548h\beta + h(66 - 6\mu)] 2i \sin(k_y h) \\
 A_{22} &= [30h^2\beta + h^2(6 - \mu)] \{2 \cos[(k_x + k_y)h] + 2 \cos[(k_x - k_y)h]\} + [120h^2\beta + h^2(18 + 2\mu)] 2 \cos(k_y h) \\
 &\quad + [-80h^2\beta + h^2(12 + 8\mu)] 2 \cos(k_x h) - 320h^2\beta + h^2(96 - 16\mu) \\
 A_{23} &= -28h^2\beta \{2 \cos[(k_x + k_y)h] - 2 \cos[(k_x - k_y)h]\} \\
 A_{31} &= [116h\beta + h(12 + 3\mu)] \{2i \sin[(k_x + k_y)h] + 2i \sin[(k_x - k_y)h]\} + [548h\beta + h(66 - 6\mu)] 2i \sin(k_y h) \\
 A_{32} &= -28h^2\beta \{2 \cos[(k_x + k_y)h] - 2 \cos[(k_x - k_y)h]\} \\
 A_{33} &= [30h^2\beta + h^2(6 - \mu)] \{2 \cos[(k_x + k_y)h] + 2 \cos[(k_x - k_y)h]\} - [80h^2\beta - h^2(12 + 8\mu)] 2 \cos(k_y h) \\
 &\quad + [120h^2\beta + h^2(18 + 2\mu)] 2 \cos(k_x h) - 320h^2\beta + h^2(96 - 16\mu). \tag{35}
 \end{aligned}$$

Equation (34) can be rewritten in matrix form as

$$[A] \begin{Bmatrix} C_1 \\ C_2 \\ C_3 \end{Bmatrix} = \begin{Bmatrix} 0 \\ 0 \\ 0 \end{Bmatrix}. \tag{36}$$

A nontrivial solution to eqn (36) exists if, and only if, the determinant of $[A]$ is zero. This leads to an equation for expressing the relationship between the frequency and wave numbers of the system. However, since there are two unknown wave numbers in the resulted equation, it is necessary to make an assumption to get the equation solved. Supposing k_x and k_y are equal to k , which is true for wave propagation in the thin plate of infinite size but an approximation for wave behaviour in the thin plate of finite size, eqn (34) is changed into the following one:

$$\begin{aligned}
 A_{11}^* C_1 + A_{12}^* C_2 + A_{13}^* C_3 &= 0 \\
 A_{21}^* C_1 + A_{22}^* C_2 + A_{23}^* C_3 &= 0 \\
 A_{31}^* C_1 + A_{32}^* C_2 + A_{33}^* C_3 &= 0, \tag{37}
 \end{aligned}$$

where

$$\begin{aligned}
 A_{11}^* &= -(394\beta + 18 + 12\mu)4 \cos^2(kh) - (2452\beta + 144 - 24\mu)4 \cos(kh) - 13816\beta + 648 - 48\mu \\
 A_{12}^* &= [116h\beta + h(12 + 3\mu)]2i \sin(2kh) + [548h\beta + h(66 - 6\mu)]2i \sin(kh) \\
 A_{13}^* &= -A_{12}^* \\
 A_{21}^* &= -\frac{1}{h} A_{12}^* \\
 A_{22}^* &= [30h\beta + h(6 - \mu)]4 \cos^2(kh) + [40h\beta + h(30 + 10\mu)]2 \cos(kh) - 320h\beta + h(96 - 16\mu) \\
 A_{23}^* &= -28h\beta [4 \cos^2(kh) - 4] \\
 A_{31}^* &= \frac{1}{h} A_{12}^* \\
 A_{32}^* &= A_{23}^* \\
 A_{33}^* &= A_{22}^*. \tag{38}
 \end{aligned}$$

Accordingly, eqn (37) is rewritten in matrix form as

$$[A^*] \begin{Bmatrix} C_1 \\ C_2 \\ C_3 \end{Bmatrix} = \begin{Bmatrix} 0 \\ 0 \\ 0 \end{Bmatrix}. \quad (39)$$

Under the condition of the determinant of $[A^*]$ being zero, a nontrivial solution to eqn (39) exists. This condition results in the following equation:

$$A_{11}^* (A_{22}^* - A_{23}^*) + \frac{2}{h} (A_{12}^*)^2 = 0. \quad (40)$$

Substituting the relevant expressions into eqn (40), the following equation yields:

$$E_1 \beta^2 + E_2 \beta + E_3 = 0, \quad (41)$$

where

$$\begin{aligned} E_1 &= 64h[1015 \cos^4(kh) - 5740 \cos^3(kh) - 20895 \cos^2(kh) + 17150 \cos(kh) + 55720] \\ E_2 &= 32h[(1080 + 545\mu) \cos^4(kh) - (435 - 1765\mu) \cos^3(kh) - (13950 + 5415\mu) \cos^2(kh) \\ &\quad - (60165 + 6095\mu) \cos(kh) - (68280 - 9200\mu)] \\ E_3 &= 32h[(90 + 45\mu + 15\mu^2) \cos^4(kh) + (225 + 135\mu - 60\mu^2) \cos^3(kh) + (135 - 675\mu + 90\mu^2) \cos^2(kh) \\ &\quad - (1305 - 765\mu + 60\mu^2) \cos(kh) + 855 - 270\mu + 15\mu^2]. \end{aligned} \quad (42)$$

From eqn (41), β and therefore, ω can be solved. In the limit, when h approaches zero, the asymptotic formula for the natural frequency of the thin plate is derived:

$$\bar{\omega} = \lim_{h \rightarrow 0} \omega = \sqrt{\frac{4D_f}{\rho}} k^2, \quad (43)$$

where $\bar{\omega}$ is viewed as the corrected natural frequency of the thin plate; k is the wave number and can be obtained from the following equation:

$$F_1 \cos^4(kh) + F_2 \cos^3(kh) + F_3 \cos^2(kh) + F_4 \cos(kh) + F_5 = 0, \quad (44)$$

where

$$\begin{aligned} F_1 &= 406\beta^2 + (216 + 109\mu)\beta + (18 + 9\mu + 3\mu^2) \\ F_2 &= -2296\beta^2 - (87 - 353\mu)\beta + (45 + 27\mu - 12\mu^2) \\ F_3 &= -8358\beta^2 - (2790 + 1083\mu)\beta + (27 - 135\mu + 18\mu^2) \\ F_4 &= 6860\beta^2 - (12033 + 1219\mu)\beta - (261 - 153\mu + 12\mu^2) \\ F_5 &= 22288\beta^2 - (13656 - 1840\mu)\beta + (171 - 54\mu + 3\mu^2). \end{aligned} \quad (45)$$

It is noted that the predicted natural frequency of a thin plate system from the finite element analysis can be corrected by simply using eqn (43). In this sense, eqn (43) and the related expressions are considered as an *a-posteriori* error estimator/corrector for the predicted natural frequency of thin plates in the finite element analysis.

If the simplest lumped mass matrix is used instead of the consistent mass matrix, eqn (43) still holds, but eqn (44) needs to be replaced by the following equation:

$$G_1 \cos^4(kh) + G_2 \cos^3(kh) + G_3 \cos^2(kh) + G_4 \cos(kh) + G_5 = 0, \quad (46)$$

where

$$\begin{aligned} G_1 &= 6 + 3\mu + \mu^2 \\ G_2 &= 15 + 9\mu - 4\mu^2 \\ G_3 &= -210(6 - \mu)\beta + 9 - 45\mu + 6\mu^2 \\ G_4 &= -1050(3 + \mu)\beta - 87 + 51\mu - 4\mu^2 \\ G_5 &= -840(6 - \mu)\beta + 57 - 18\mu + \mu^2. \end{aligned} \tag{47}$$

4. AN APPLICATION EXAMPLE

In theory, the error estimator/corrector presented in the last section is an exact one for estimating/correcting the natural frequency error of an infinite thin plate, since the wave number in the x -direction is equal to that in the y -direction for a thin plate of infinite size. In practice, however, when the thin plate is of finite size, the wave number in the x -direction is generally different from that in the y -direction. In this case, the present error estimator/corrector can only be used as an approximate one for the error estimate/correction of the natural frequency of finite thin plates in the finite element analysis. The free-vibration of an elastic, isotropic square thin plate with four sides simply supported is selected as an application example to assess the present error estimator/corrector when it is used for estimating/correcting the natural frequency error of a thin plate of finite size.

As shown in Fig. 3, an elastic, isotropic square thin plate of side length, h , is discretized using one thin plate element only so that it forms a critical problem since the minimum element number is used in the analysis. The analytical solution is available [1] for the natural frequency of the square thin plate with four sides simply supported. Comparing the analytical solution with the finite element solution before and after correction, the usefulness of the present error estimator/corrector can be illustrated for estimating/correcting the natural frequency error of a thin plate of finite size in the finite element analysis. For simplicity, the following parameters are used in the calculation: $\rho = 1000 \text{ kg m}^{-3}$, $D_f = 1 \text{ kN-m}$, $\mu = 0.3$ and $h = 1 \text{ m}$.

Table 1 shows the numerical error comparison between the uncorrected and corrected natural frequencies of the thin square plate. It can be seen from these numerical results that the error of numerical solutions for the natural frequencies of the thin square plate after correction is considerably reduced, compared with that before correction. This fact demonstrates that the present error estimator/corrector can lead to a considerable, although not perfect, improvement on the predicted natural frequency from the finite element analysis when

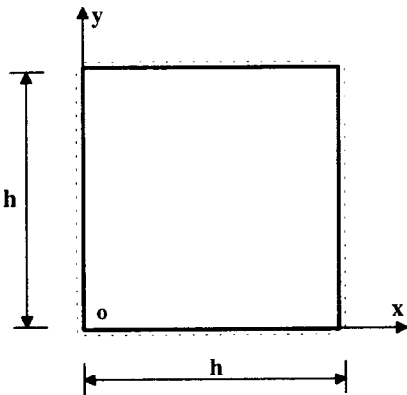


Fig. 3. Finite element model of a square thin plate.

Table 1. Numerical error comparison between the uncorrected and corrected natural frequencies

Frequency	Exact solution (rad s ⁻¹)	Numerical solution before correction	Error before correction	Numerical solution after correction	Error after correction
ω_2	49.35	18.19	63.14%	39.7	19.55%
ω_3	78.96	26.83	66.02%	90.04	14.03%
ω_4	98.7	45.56	53.84%	96.14	2.57%

the thin plate is of finite size. However, it is expected that further refinement on the present error estimator/corrector might be achieved if the following alternatives are considered: (1) a conforming thin plate bending element is used instead of the non-conforming one; (2) an optimal value for the ratio of the wave number in the x -direction to that in the y -direction is used to replace the assumption employed in this study.

5. CONCLUSIONS

In order to develop an *a-posteriori* error estimator/corrector for the predicted natural frequencies of thin plate vibration problems from the finite element analysis, explicit expressions for the stiffness matrix and mass matrix of a thin plate bending element are needed and therefore derived in this paper. Based on the idea that a discretized finite element system approaches a continuous system when the finite element size approaches zero, the asymptotic formula is presented to describe the asymptotic behaviour of the predicted natural frequency of a thin plate system in the finite element analysis. This asymptotic formula can be used as an exact *a-posteriori* error estimator/corrector for estimating/correcting the natural frequency error when the thin plate is of infinite size but as an approximate one when the thin plate is of finite size. Further refinement on the present error estimator/corrector might be achieved if some alternatives are considered.

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